AI Powered Intelligent Hospital Patient Flow Management System

RESULTS

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1. Measurement Conditions

Measurements were taken on Windows 11 Pro 64-bit, Intel Core i5-12500 (6 cores, 12 logical processors), 15.7 GB RAM and Python 3.11.9. Models were evaluated on deterministic synthetic test partitions. API timings are 15 sequential loopback requests per read endpoint against Uvicorn, FastAPI and SQLite. The screenshots use a synthetic demonstration database and Google Chrome at 1440 x 900.

Table 1. Measured model results.

Model	Evidence	Metric	Value
Wait-time regressor	12,000 synthetic rows	R-squared	0.951041
Wait-time regressor	1,800-row test	MAE	5.982 min
Wait-time regressor	1,800-row test	RMSE	8.580 min
Triage classifier	2,250-row test	Accuracy	0.9680
Triage classifier	2,250-row test	Macro F1	0.898475
Triage classifier	100 critical cases	Critical recall	0.63

2. Model Performance

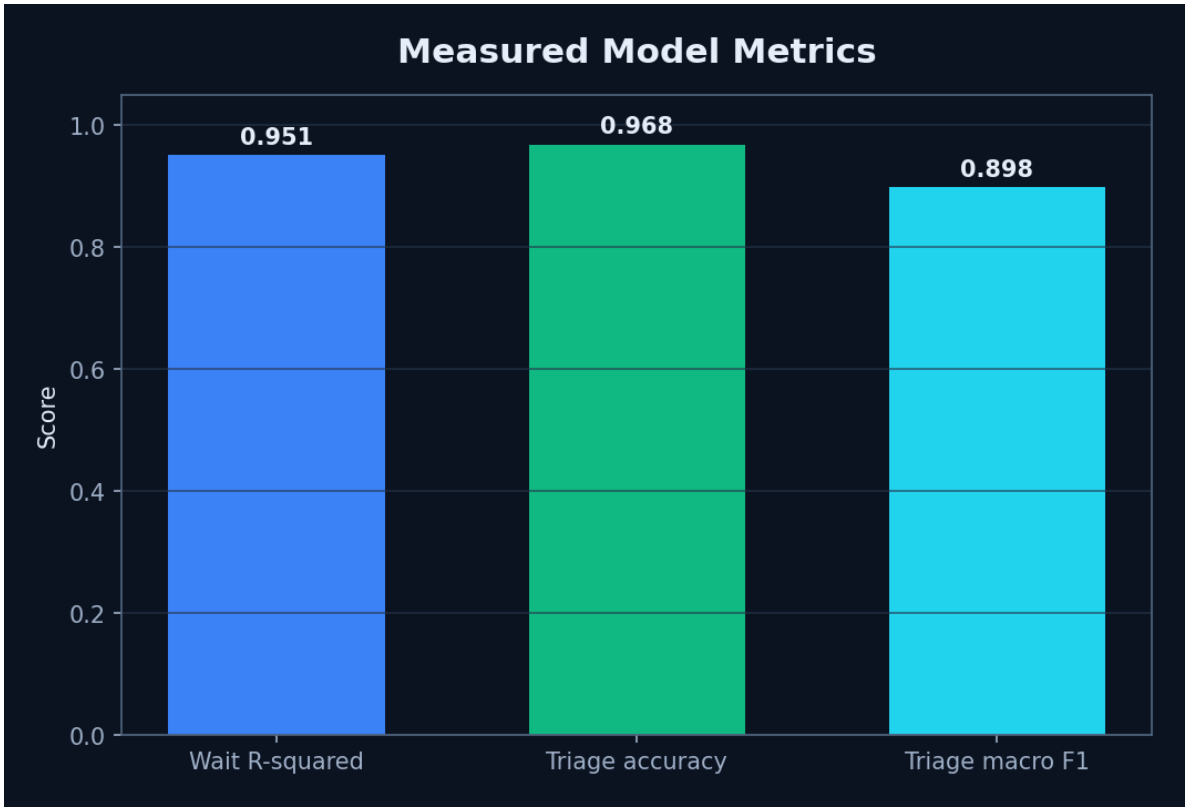


Figure 1. Measured model metrics.

The wait model reconstructs its designed synthetic relationship with R-squared 0.951041. Its MAE of 5.982 minutes and RMSE of 8.580 minutes show that larger misses remain. The triage model reached 96.80% accuracy and 0.898475 macro F1, while its 63% critical recall is the decisive safety limitation.

3. Triage Error Analysis

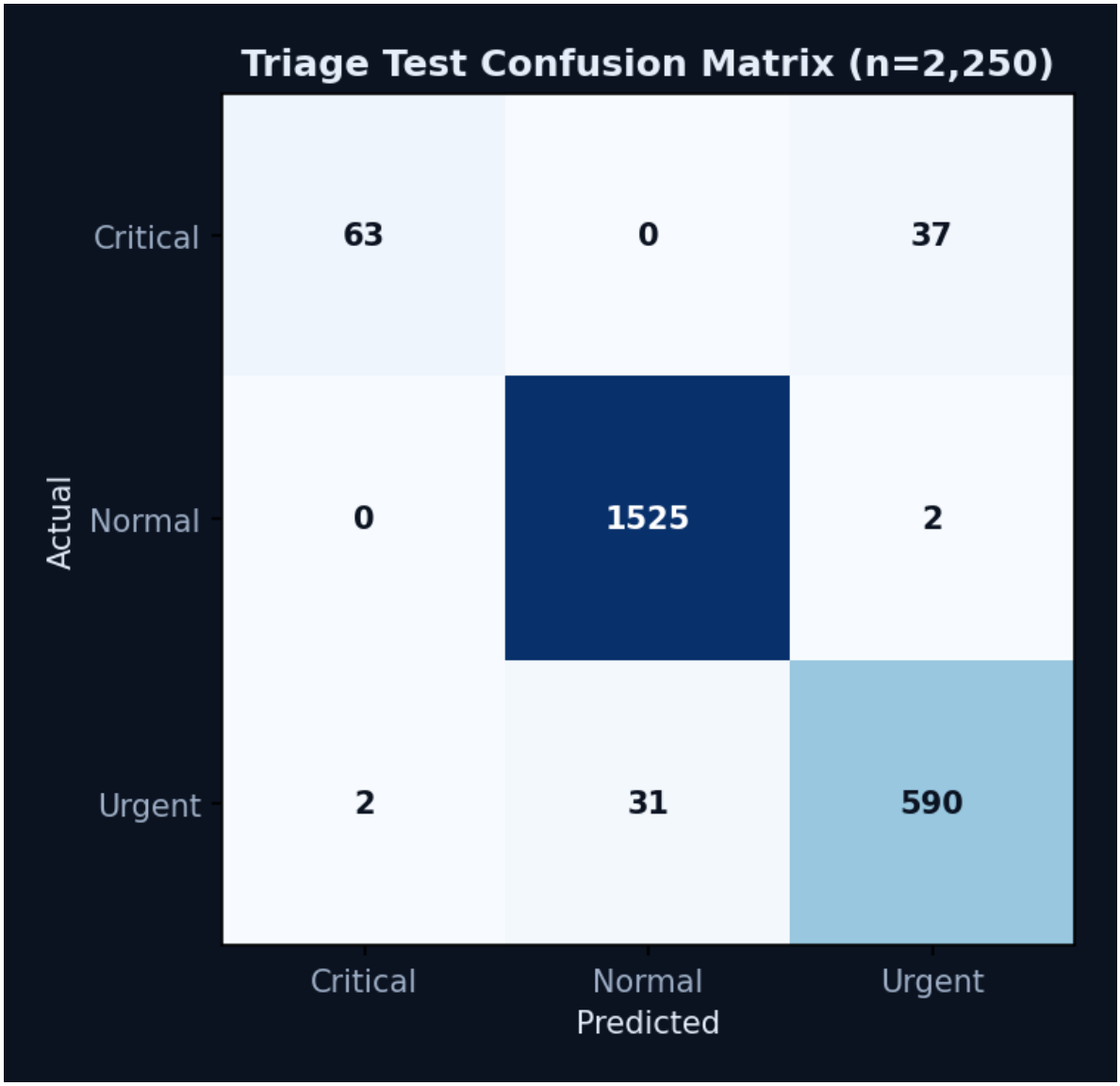


Figure 2. Triage confusion matrix.

Thirty-seven of 100 actual critical examples were classified as urgent, while two urgent cases were classified as critical and 31 urgent cases as normal. The classifier therefore requires clinician override, threshold/cost-sensitive investigation and external validation before any clinical use.

4. API and Build Results

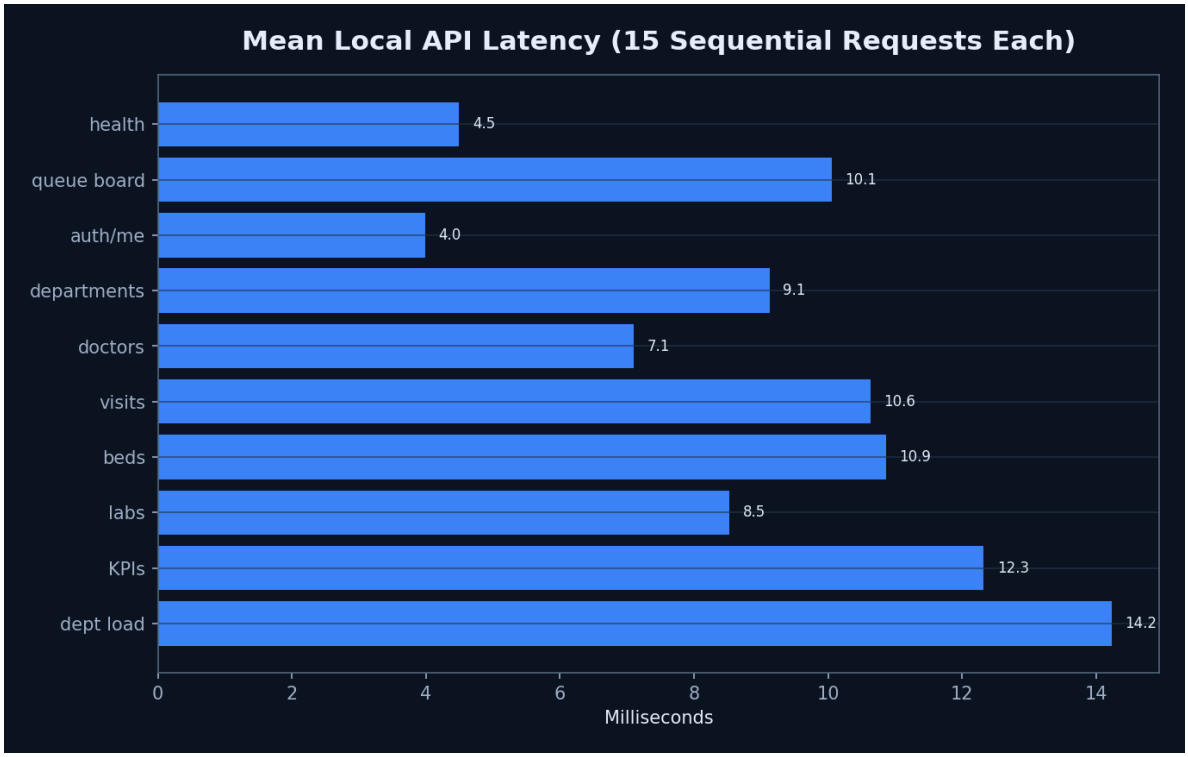


Figure 3. Mean local API latency.

All 150 measured read requests returned HTTP 200. Mean endpoint times ranged from 3.985 ms for /api/auth/me to 14.231 ms for department load. One WebSocket connection completed in 46.657 ms. These are local sequential measurements and not a capacity claim.

Frontend lint exited zero with 14 warnings. The production build exited zero after 3,573 modules and 1.73 seconds, with warnings for two JavaScript chunks larger than 500 kB.

Public waiting board

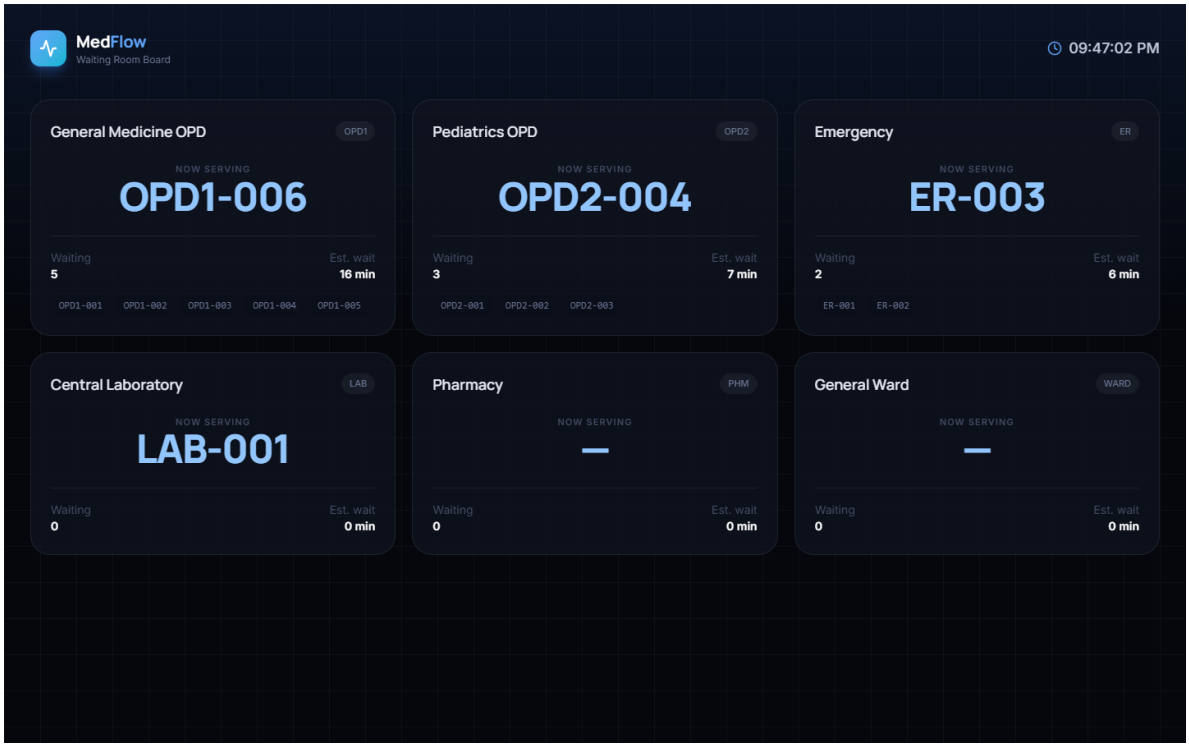


Figure 4. Public waiting board

Live department cards show now-serving tokens, waiting counts, upcoming tokens and mean predicted waits.

Administrator command centre

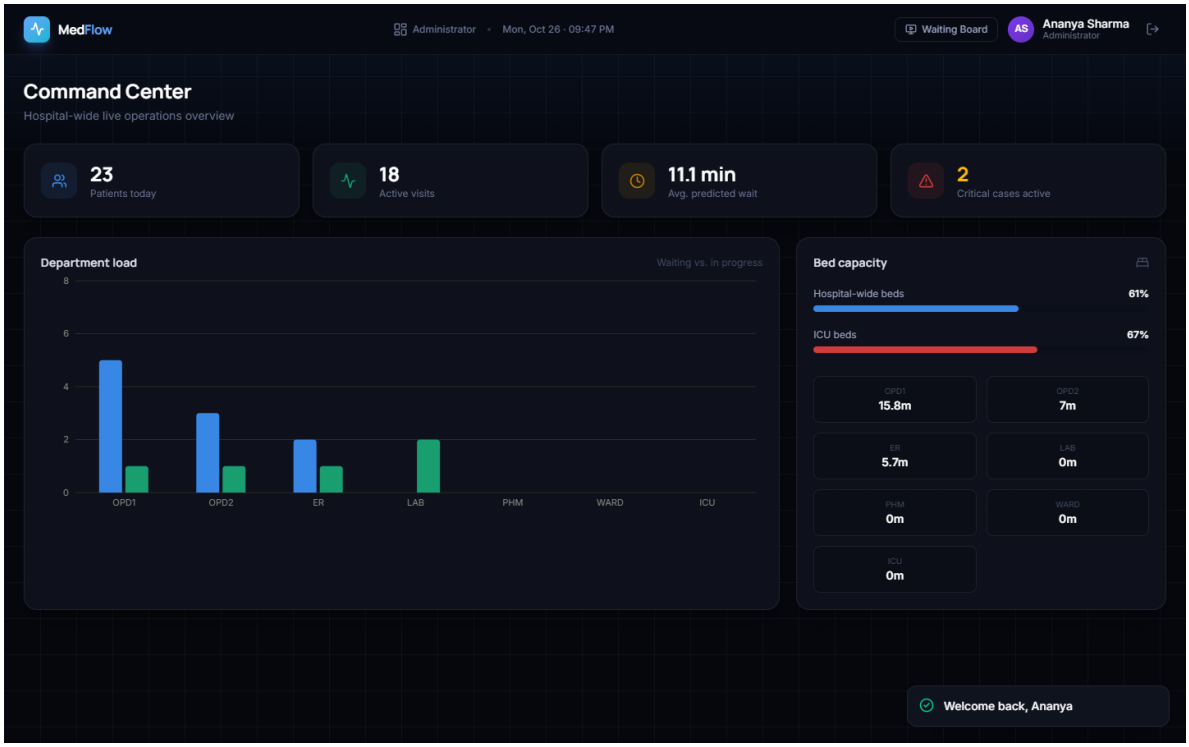


Figure 5. Administrator command centre

The demonstration snapshot shows 23 patients today, 18 active visits, 11.1-minute predicted wait, two active critical cases and capacity.

Reception dashboard

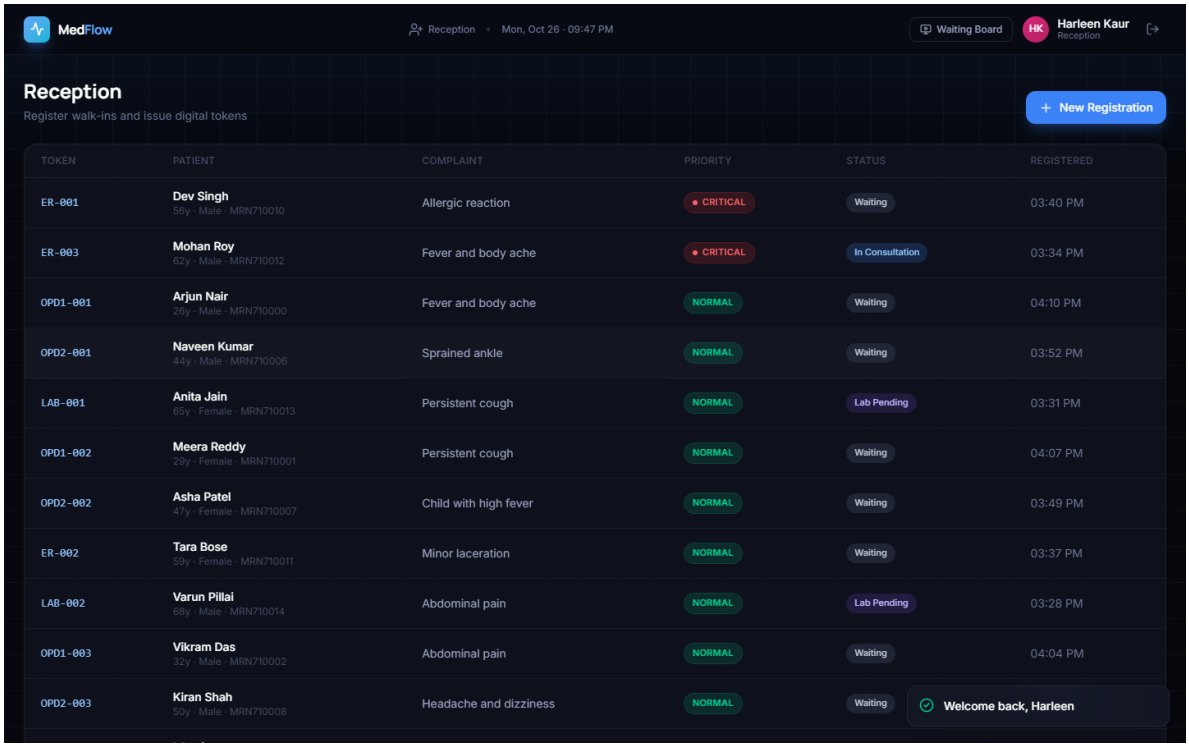


Figure 6. Reception dashboard

Token, patient, complaint, priority, state and registration time are visible in one table.

Registration and triage input

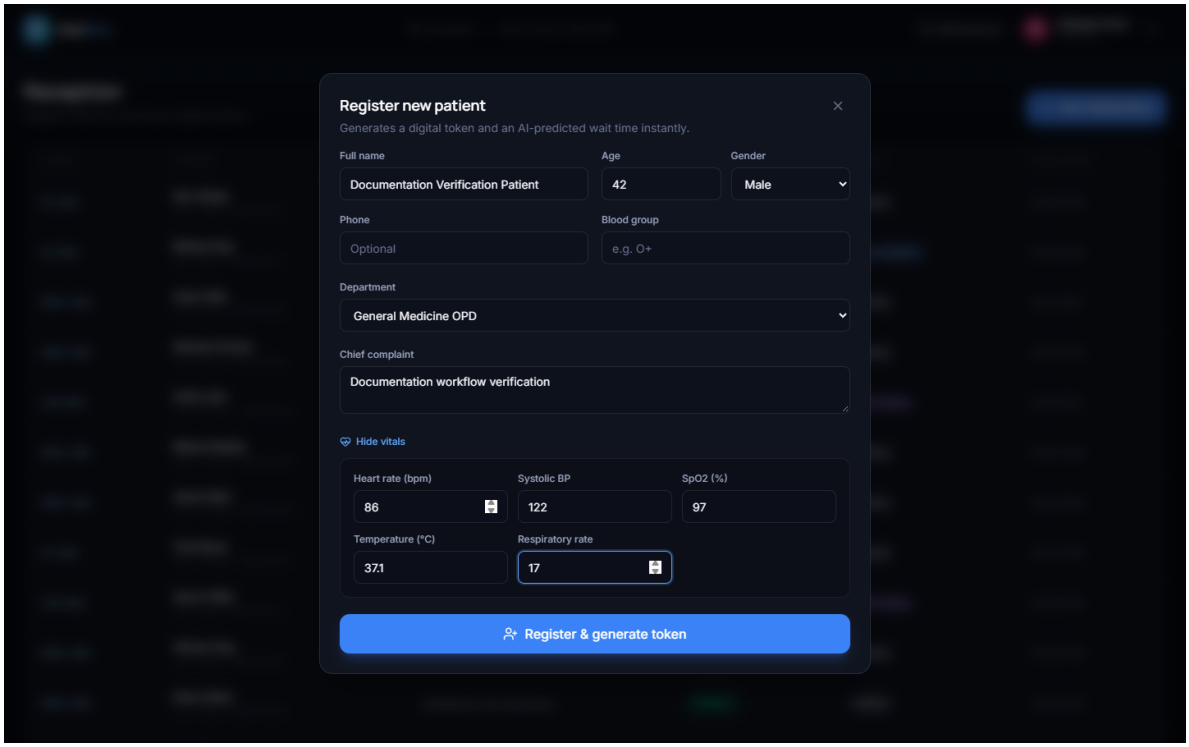


Figure 7. Registration and triage input

The real form accepts patient identity, department, complaint and optional vitals.

Doctor dashboard

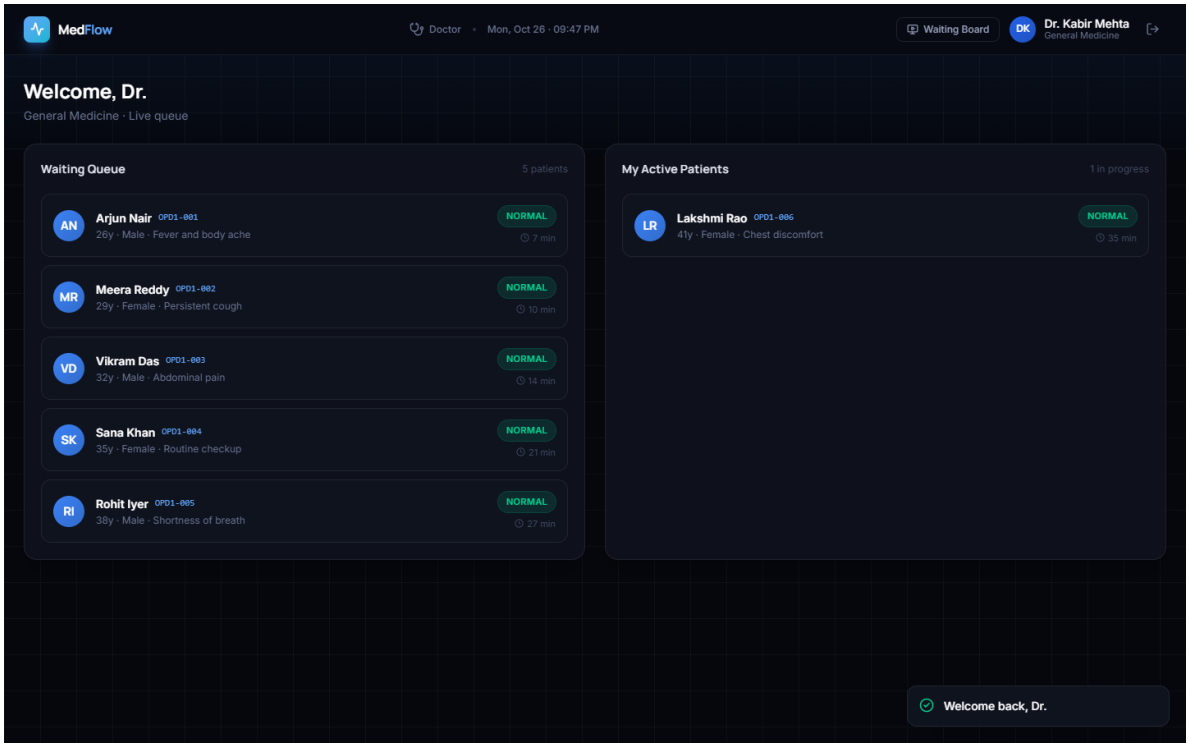


Figure 8. Doctor dashboard

Waiting and assigned patients are separated while retaining priority and predicted wait.

Visit detail

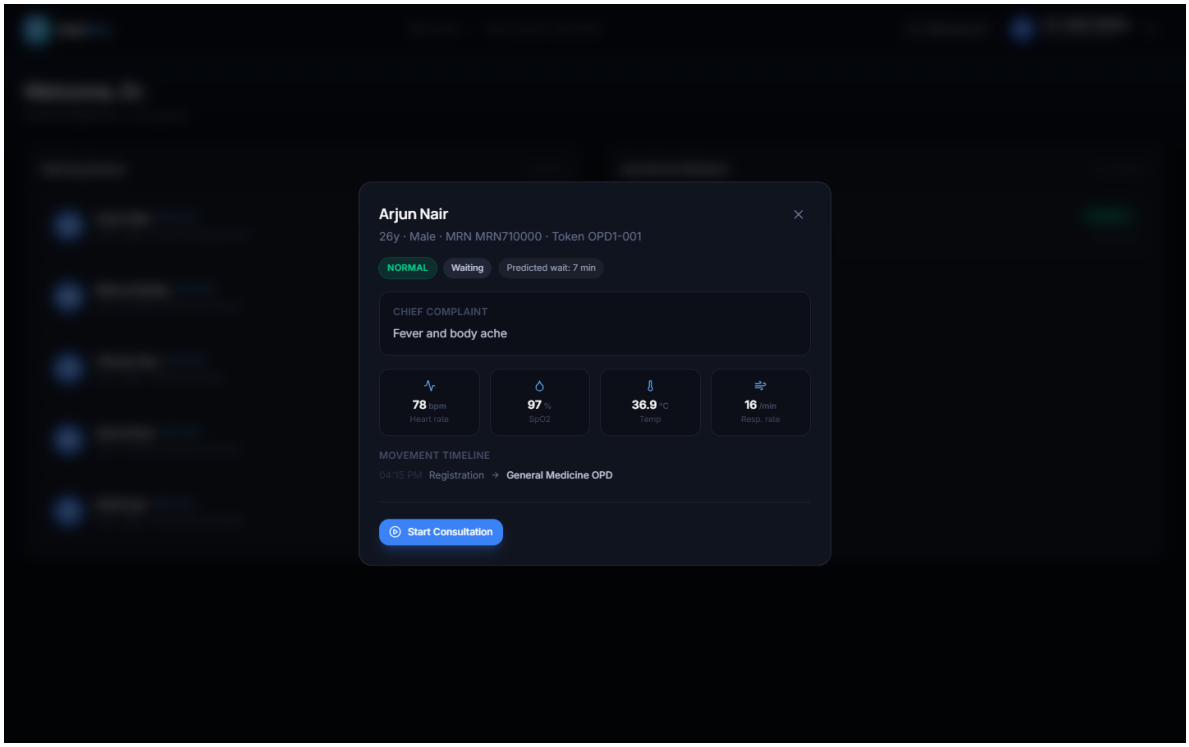


Figure 9. Visit detail

Vitals, complaint, priority, wait and movement history support the consultation decision.

Laboratory order modal

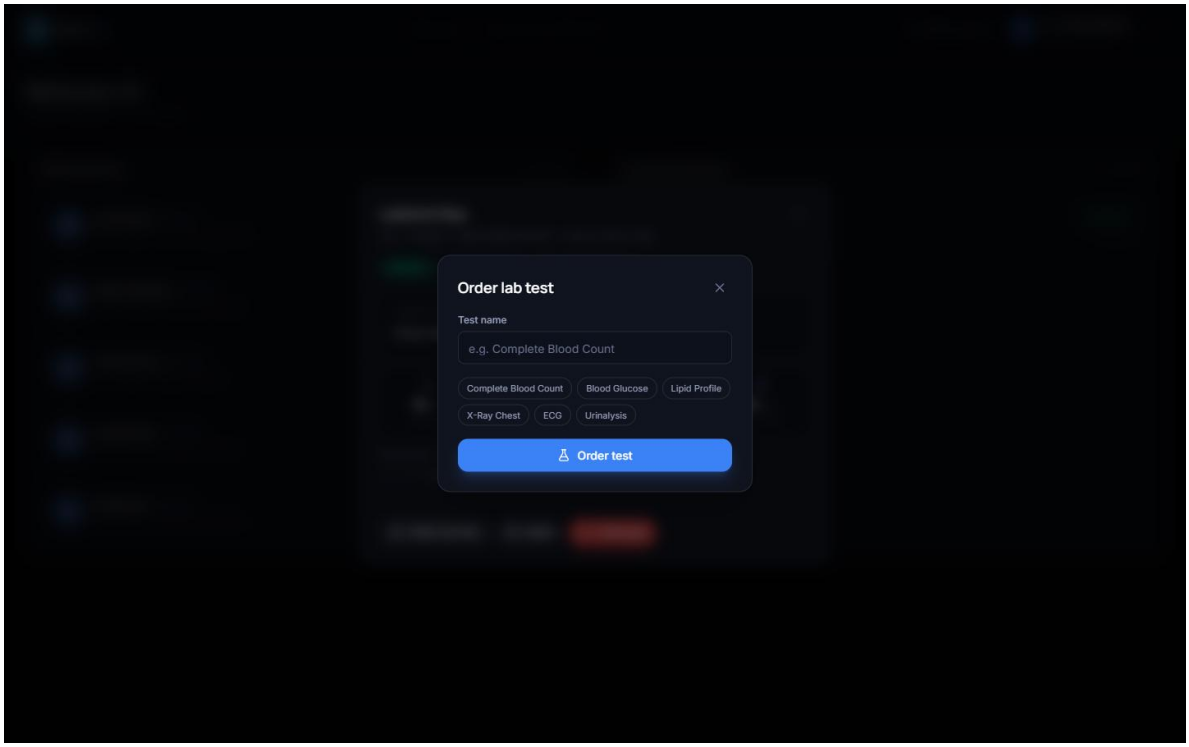


Figure 10. Laboratory order modal

Doctors may enter a named test or choose a frequent test.

Bed admission modal

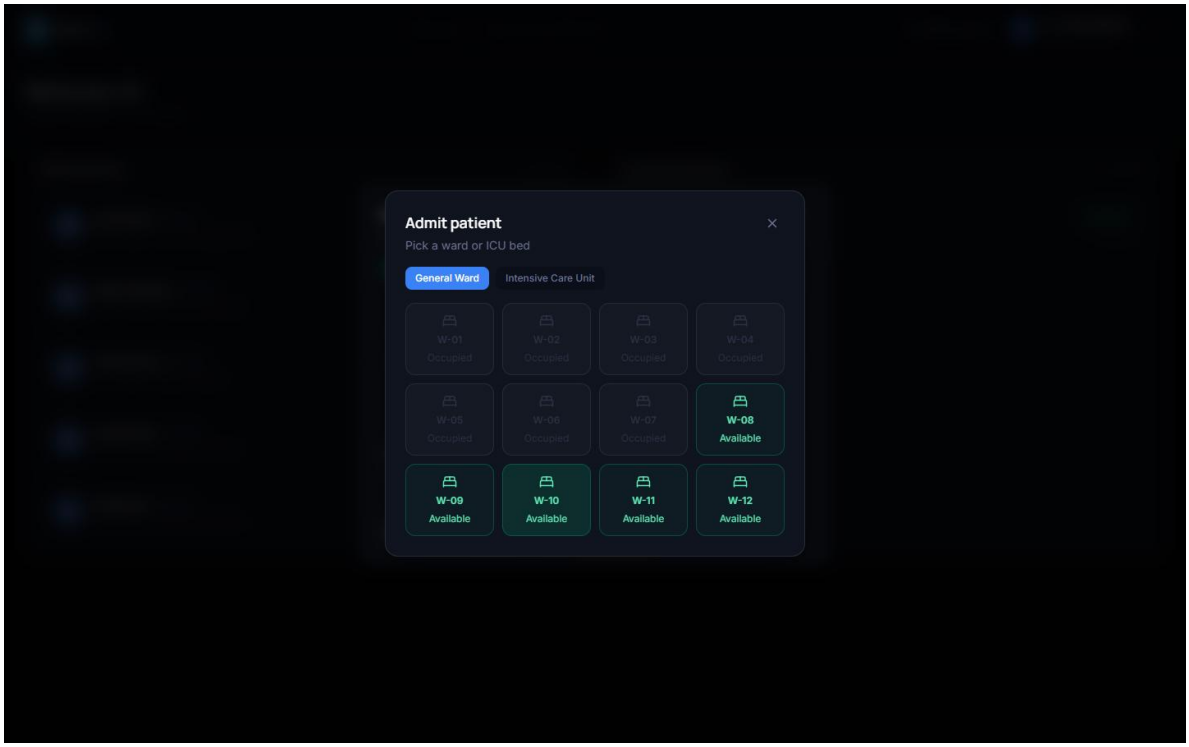


Figure 11. Bed admission modal

Occupied beds are disabled and available ward or ICU beds are selectable.

General Ward bed board

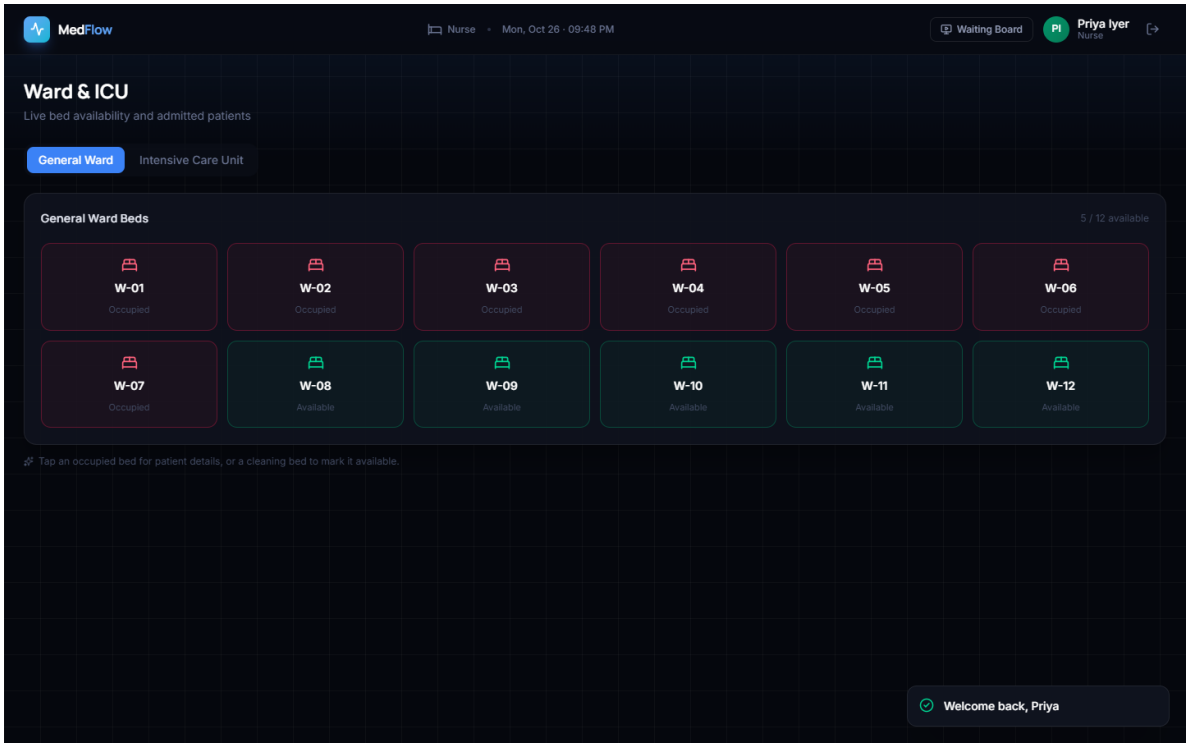


Figure 12. General Ward bed board

The synthetic snapshot contains 12 ward beds: seven occupied and five available.

ICU bed board

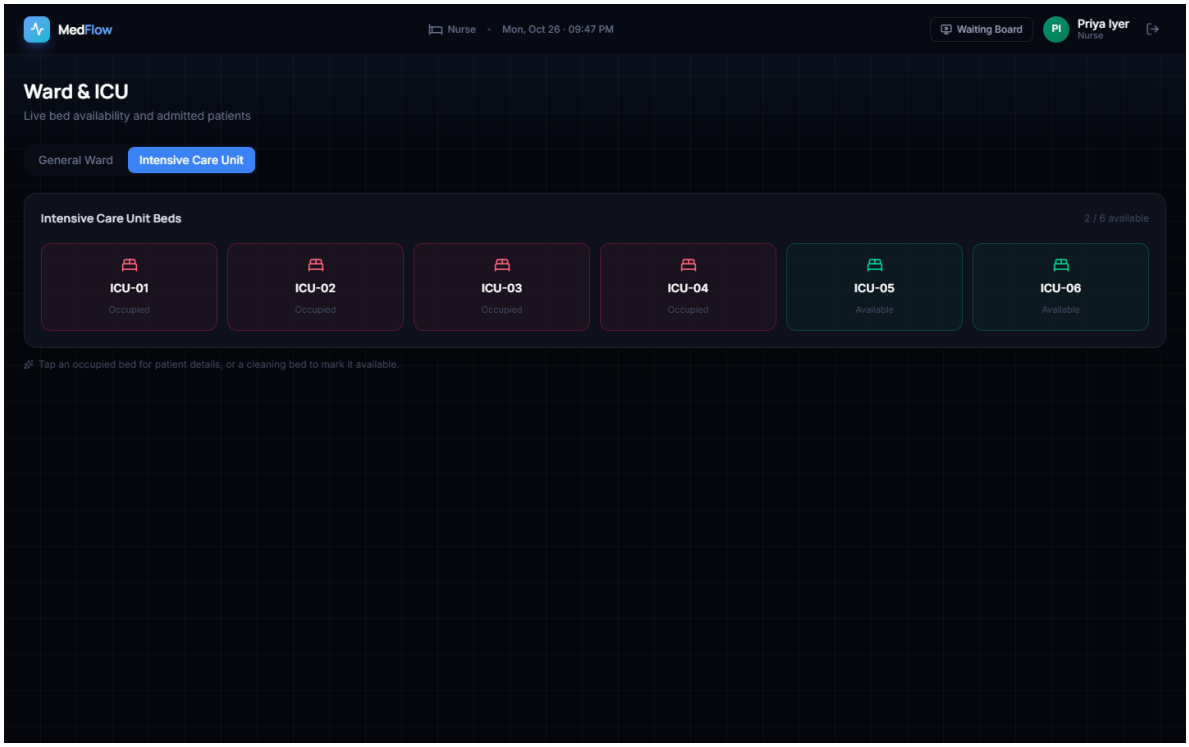


Figure 13. ICU bed board

The synthetic snapshot contains six ICU beds: four occupied and two available.

Laboratory workflow

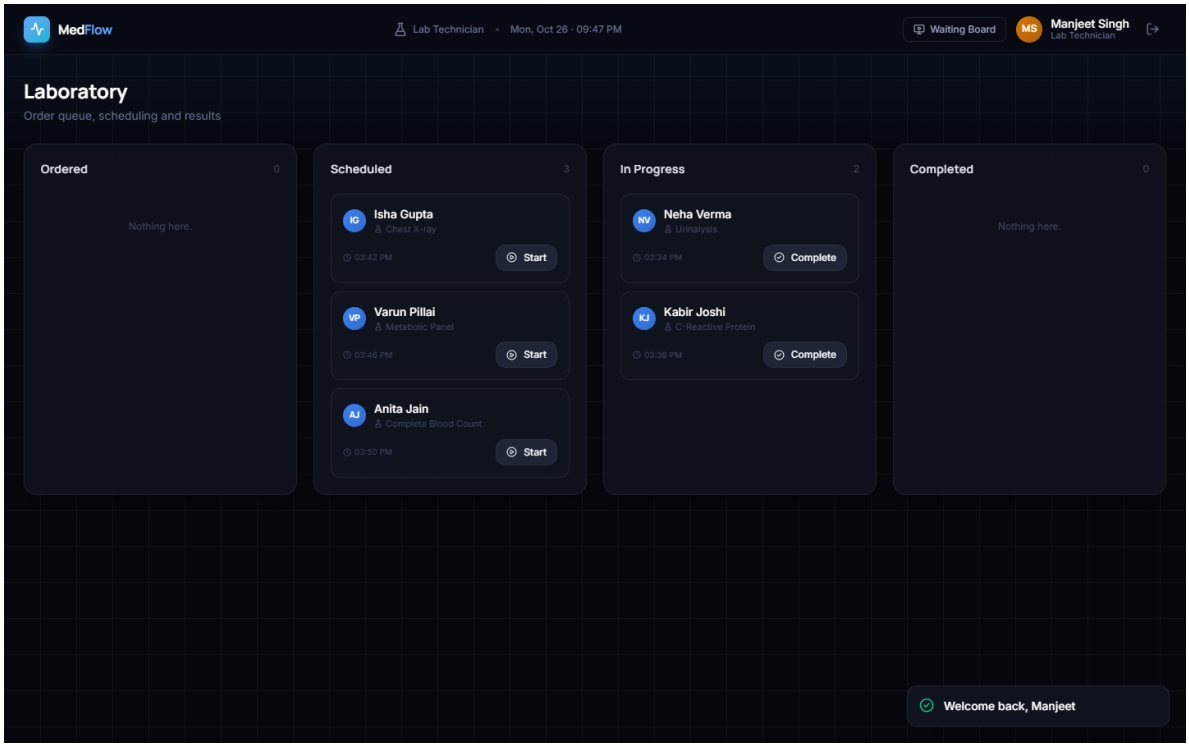


Figure 14. Laboratory workflow

Five tests appear across scheduled and in-progress columns with valid actions.

16. Objectives versus Outcomes

Table 2. Objectives and outcomes.

Objective	Outcome	Evidence
Digital tokens	Achieved	Token allocation and queue board
Movement tracking	Partially achieved	Registration and explicit department changes logged
Lab scheduling	Achieved as workflow state	Five populated tests; no optimisation study
Live bed/ICU monitoring	Achieved for state visibility	Ward and ICU boards
AI wait and priority	Prototype achieved	Measured synthetic-model results
Role dashboards	Achieved	Five staff roles plus public board

17. Success and Failure Cases

Successful cases include authenticated access, priority-first queue ordering, coherent token display, model inference, laboratory-state display, bed-capacity display and live dashboard refresh. The measured endpoints all returned HTTP 200 under the defined smoke conditions.

Observed limits include 37 critical-to-urgent triage errors in the synthetic test, the nurse page requiring an initial tab click, incomplete coupling between laboratory and visit states, seeded occupied beds without visit links, large frontend chunks and absence of automated tests. No unsupported wait-reduction claim is made.

18. Results Limitations

Both model datasets are synthetic and generated from designed distributions; they are not real patient records.

The measured model test partitions come from the same deterministic generators as training and are not independent clinical validation.

No real hospital deployment, clinician study, user-acceptance study, security audit, accessibility audit, or concurrent-load test was performed.

The 96.80% triage accuracy must be interpreted with the 63% critical-class recall; accuracy alone overstates safety-critical performance.

SQLite, a development JWT secret fallback, an unauthenticated WebSocket, and in-process broadcasting are unsuitable for production without hardening.

The landing-page claims of 42% wait reduction, 214 patients, and six departments are hard-coded marketing copy and are excluded from results.